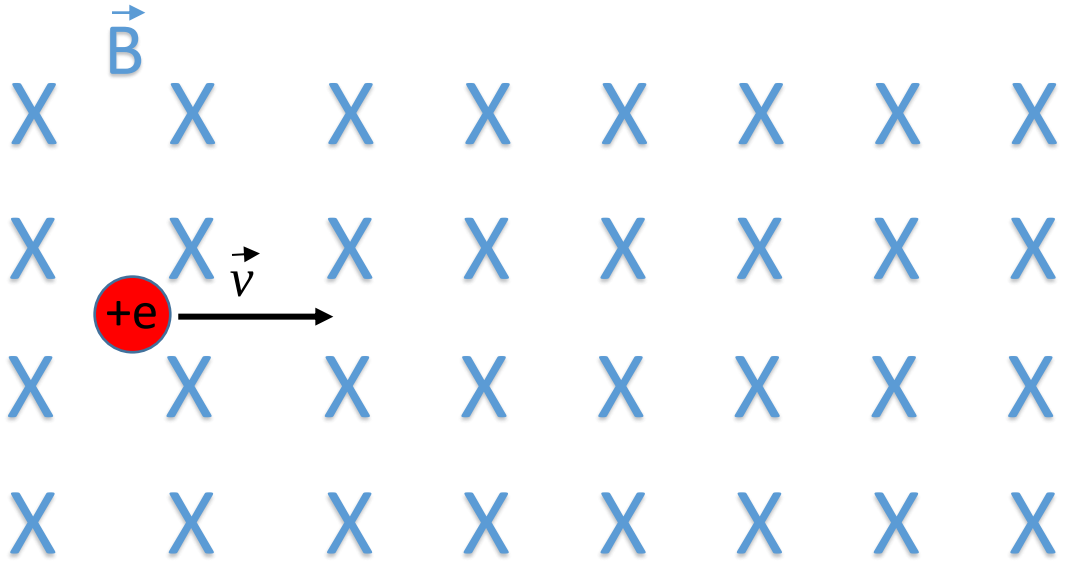
Exercise 1.



A proton moves to the right in a homogeneous magnetic field that points into the plane as shown in the figure. The proton is not affected by forces other than the magnetic one. Which of the following statements are correct at the instant shown in the figure?

A) The proton's acceleration is directed upwards

B) The proton's acceleration is directed downwards

C) The proton's acceleration is directed into the plane

D) The proton's acceleration is directed out of the plane

E) The acceleration of the proton is zero

F) The kinetic energy of the proton increases

G) The kinetic energy of the proton decreases

H) The kinetic energy of the proton does not change

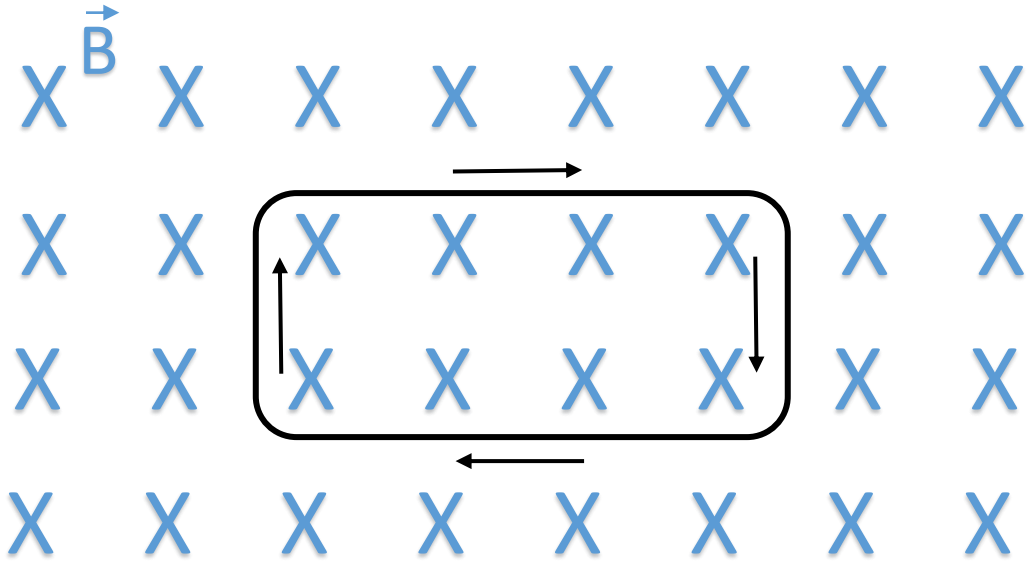
I) Don't know

Exercise 2.

In a ’synchrotron’ protons with a kinetic energy of $2.0 \cdot 10^6$ eV are held in a circular path with a radius of 50 cm by means of a magnetic field perpendicular to the path. What is the required magnetic field strength?

A) 0.20 T
B) 0.29 T
C) 0.41 T
D) 0.53 T
E) 0.69 T
F) Don’t know

Exercise 3.



A current I =10.0 A circulates clockwise in a rectangular loop as shown in the figure. The loop is situated in a homogeneous magnetic field of magnitude B=0.154 T pointing into the plane of the figure. The width of the loop is denoted a=20.0 cm, and its height b=10.0 cm.

a) Calculate the magnitude and direction of the magnetic force on each of the sides of the rectangle (ignore the forces on the rounded corners).

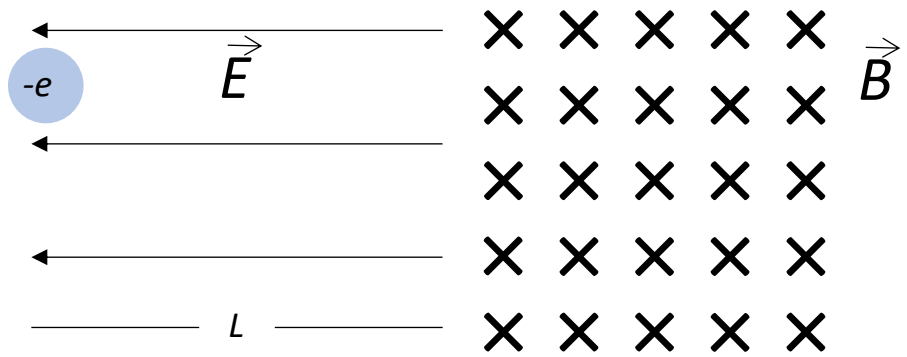
b) Calculate the magnetic moment of the current loop (magnitude and direction).

c) Calculate the total torque on the current loop around its center directly from the results in a). Is the result consistent with Eq. 11.22 in the textbook?

d) Calculate the work required to turn the current loop 180 degrees around a lateral axis so that the current flow is counterclockwise in the final state.

Hint for d): Eq. 11.23 in the textbook.

Exercise 4.



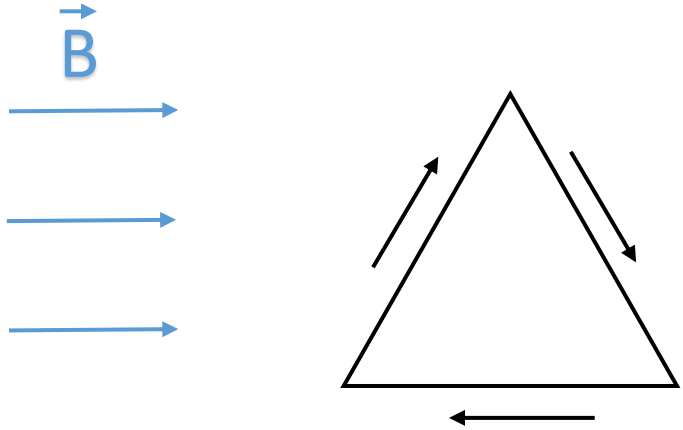
An electron is accelerated in a homogeneous electric field whose field strength is *E* over a distance *L* as shown in the figure. Subsequently the electron moves into a homogeneous magnetic field of strength *B* pointing into the plane of the figure. Let *e* denote the elementary charge, and *m* the mass of the electron.

a) Determine the speed of the electron after its acceleration in the *E*-field.

b) Determine an expression for the radius *R* of the circular trajectory that the electron will follow when it enters the magnetic field.

c) Determine the direction of the electrons circular movement (clockwise/counterclockwise).

Exercise 5.

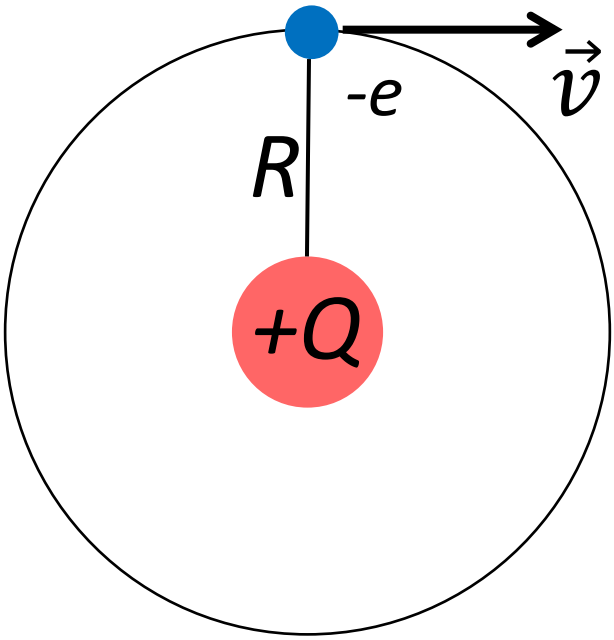


A current loop shaped like an equilateral triangle with side length a=30.0 cm is situated in a homogeneous magnetic field of strength B=1.20 T, pointing left-to-right as shown in the figure. The loop carries a current I=5.00 A.

a) Determine the magnitude and direction of the torque on the loop.

b) What work must be done on the loop to turn it into the orientation where its potential energy is minimized?

Exercise 6.



An electron is doing uniform circular movement around a spherically symmetric charge distribution having a total charge of Q=0.1 nC as shown in the figure. The entire path of the electron is within a uniform magnetic field of strength B perpendicular to the plane of the figure. The speed of the electron in the circular path is 10^6 m/s, and the radius of the path is R=0.04 m.

a) Set up expressions for the magnitude of electric and magnetic forces on the electron.

b) What will be the direction of the magnetic force if the B-field points out of, or into, the plane? Note that the direction of circulation is given in the figure.

c) Write up Newton's 2. law for the circular movement.

d) Solve the equation for B, and determine the strength and direction of the magnetic field.